HELMHOLTZ AI

Introduction to Explainable AI

Helmholtz AI @ HIDA Data Science Courses
24.-25.11.2026

Helmholtz AI Consultants

Who are we and what is our mission?

WHAT IS OUR MISSION?



★ Maximise research impact by democratising access to AI



★ We empower health researchers with tailored AI solutions, and strive to translate them into reusable tools for the community.

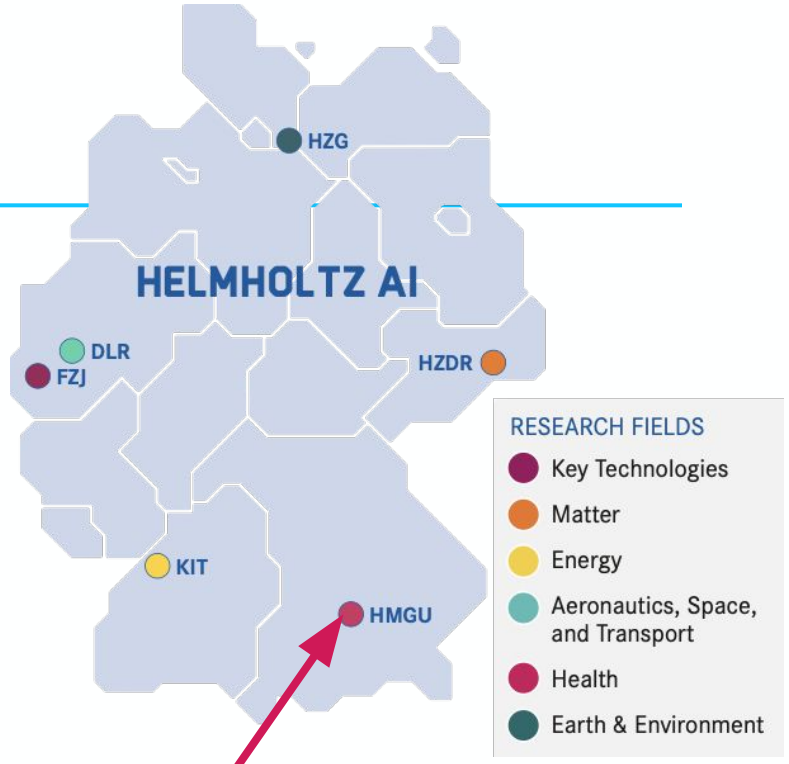
HOW DO WE DO THAT?

★ Short- to mid-term scientific collaboration
★ Free of Charge
★ Easy Application





https://voucher-system.helmholtz.ai/





Ask your questions in the chat!

Agenda

Introduction to eXplainable AI (XAI)

Day 1: XAI for Random Forests

- Model-agnostic methods: Perm. Feature Importance, LIME, SHAP
- Model-specific methods: Forest-Guided Clustering
- Comparison of XAI methods for Random Forest models

Day 2: XAI for CNNs

- Model-agnostic methods: LIME, SHAP
- Model-specific methods: Grad-CAM
- Comparison of XAI methods for CNN models

Wrap-Up

GitHub Repository:
https://tinyurl.com/4y46r666

Introduction to eXplainable AI (XAI)

Terminology

Interpretability or Explainability?

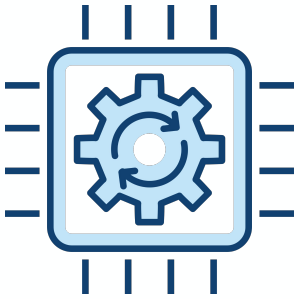


Introduction to eXplainable AI (XAI)

Terminology

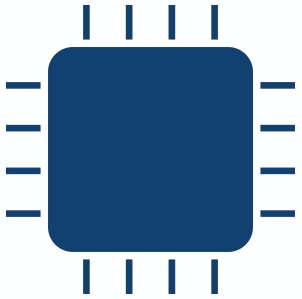
Interpretability

The degree to which a human can understand the internal mechanics of a model without external tools, focussing on the transparency of the model itself.



Explainability

The extent to which the internal mechanics of a machine learning model can be explained in human terms, often using post-hoc methods, focussing on insights into the behavior of black-box models.

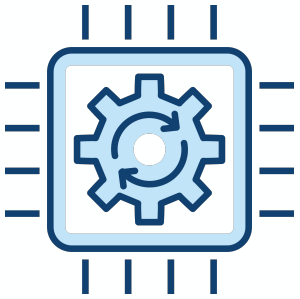


Introduction to eXplainable AI (XAI)

Terminology

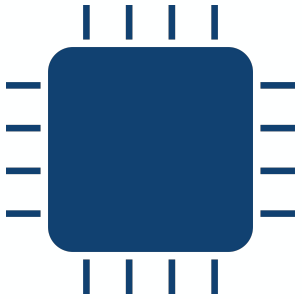
Interpretability

The degree to which a human can understand the internal mechanics of a model without external tools, focussing on the transparency of the model itself.



Explainability

The extent to which the internal mechanics of a machine learning model can be explained in human terms, often using post-hoc methods, focussing on insights into the behavior of black-box models.

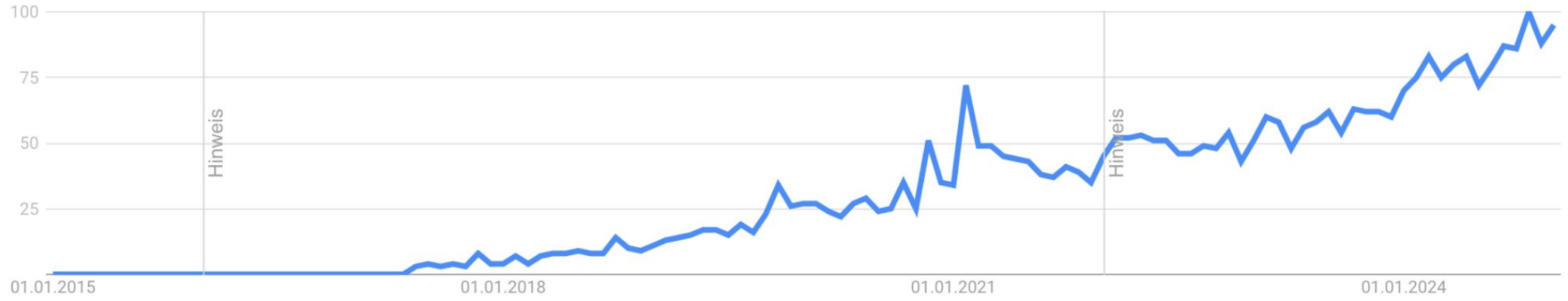


In this course, we will focus only on eXplainable Artificial Intelligence (XAI).

Introduction to eXplainable AI (XAI)

Why is explainability important?

Google Trends Popularity Index of the term "Explainable AI" over the last ten years (2015–2025)



slido



Why is explainability important?

ⓘ Start presenting to display the poll results on this slide.

Introduction to eXplainable AI (XAI)

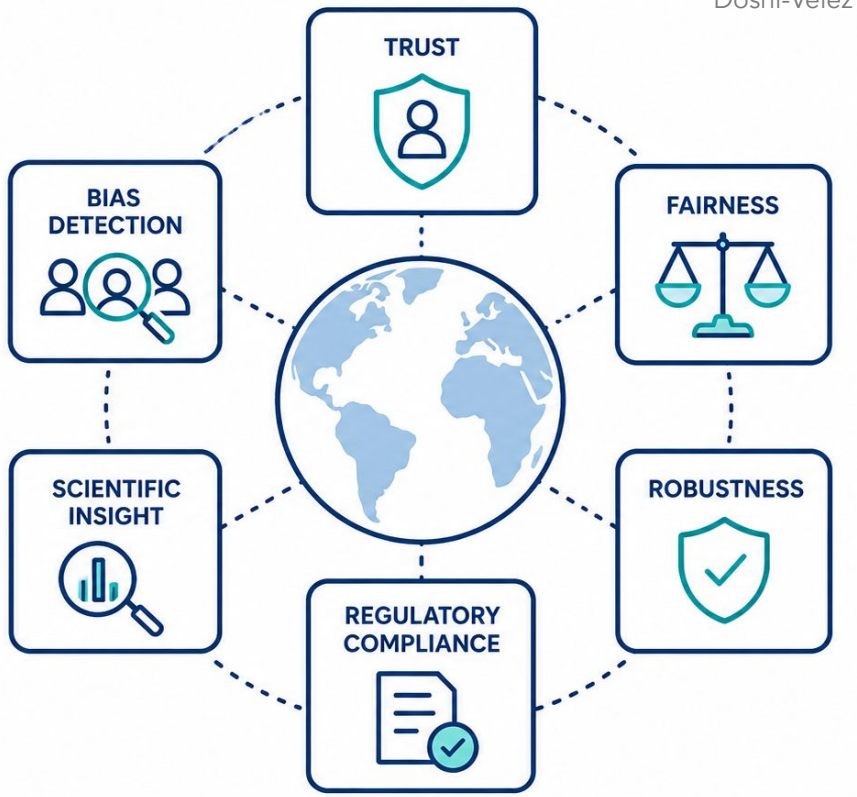
Why is explainability important?

The problem is that a single metric …



… is an incomplete description of most real world tasks.

Doshi-Velez et al., 2017



Introduction to eXplainable AI (XAI)

Why is explainability important? For technological acceptance!



Introduction to eXplainable AI (XAI)

Why is explainability important? To avoid ethical issues!

NEWS | 24 October 2019 | Update 26 October 2019

Millions of black people affected by racial bias in health-care algorithms

Study reveals rampant racism in decision-making software used by US hospitals – and highlights ways to correct it.

Heidi Ledford



Introduction to eXplainable AI (XAI)

Why is explainability important? For knowledge creation!

What Does Deep Learning See? Insights From a Classifier Trained to Predict Contrast Enhancement Phase From CT Images

Kenneth A. Philbrick[1]
Kotaro Yoshida
Dai Inoue
Zeynettin Akkus
Timothy L. Kline
Alexander D. Weston
Panagiotis Korfiatis
Naoki Takahashi
Bradley J. Erickson

OBJECTIVE. Deep learning has shown great promise for improving medical image classification tasks. However, knowing what aspects of an image the deep learning system uses or, in a manner of speaking, sees to make its prediction is difficult.

MATERIALS AND METHODS. Within a radiologic imaging context, we investigated the utility of methods designed to identify features within images on which deep learning activates. In this study, we developed a classifier to identify contrast enhancement phase from whole-slice CT data. We then used this classifier as an easily interpretable system to explore the utility of class activation map (CAMs), gradient-weighted class activation maps (Grad-CAMs), saliency maps, guided backpropagation maps, and the saliency activation map, a novel map reported here, to identify image features the model used when performing prediction.

RESULTS. All techniques identified voxels within imaging that the classifier used. SAMs had greater specificity than did guided backpropagation maps, CAMs, and Grad-CAMs at identifying voxels within imaging that the model used to perform prediction. At shallow network layers, SAMs had greater specificity than Grad-CAMs at identifying input voxels that the layers within the model used to perform prediction.

CONCLUSION. As a whole, voxel-level visualizations and visualizations of the imaging features that activate shallow network layers are powerful techniques to identify features that deep learning models use when performing prediction.

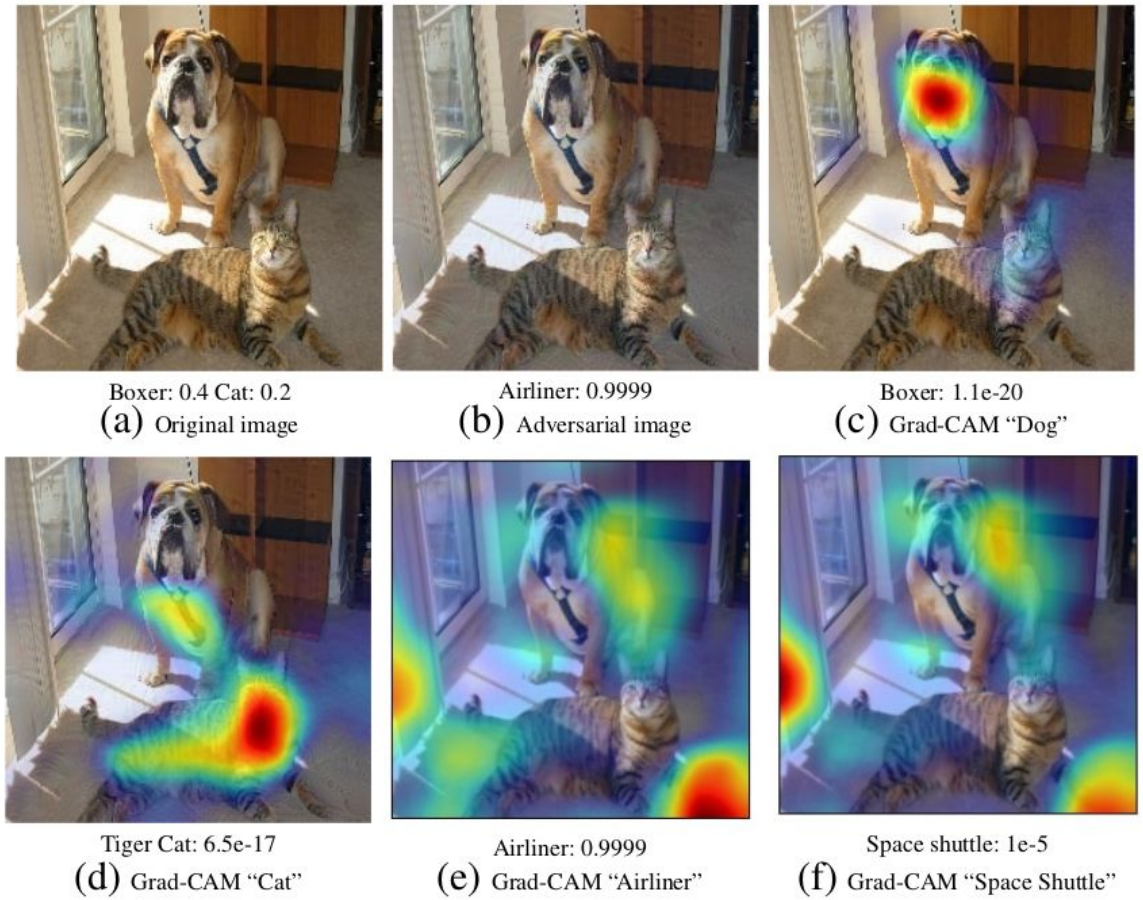
Introduction to eXplainable AI (XAI)

Why is explainability important? To meet regulatory requirements!



Introduction to eXplainable AI (XAI)

Why is explainability important? To Support Robustness Analysis!



https://campusai.github.io/_papers/Grad-CAM/adversarial.png

Home › Artificial Intelligence and Soft Computing › Conference paper

Explainable AI for Inspecting Adversarial Attacks on Deep Neural Networks

Zuzanna Klawikowska, Agnieszka Mikołajczyk & Michał Grochowski

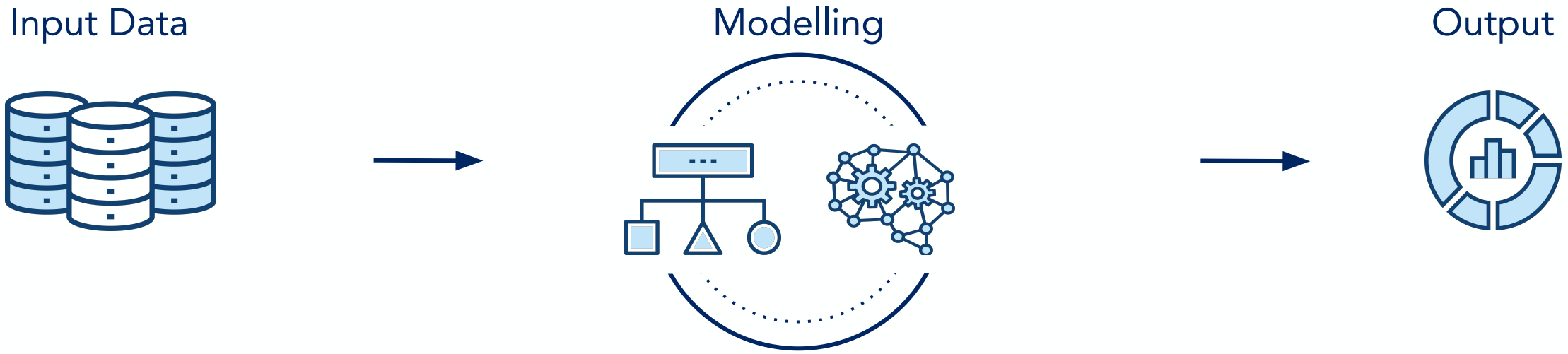
Conference paper | First Online: 07 October 2020

2252 Accesses | 1 Citations

Part of the Lecture Notes in Computer Science book series (LNAI,volume 12415)

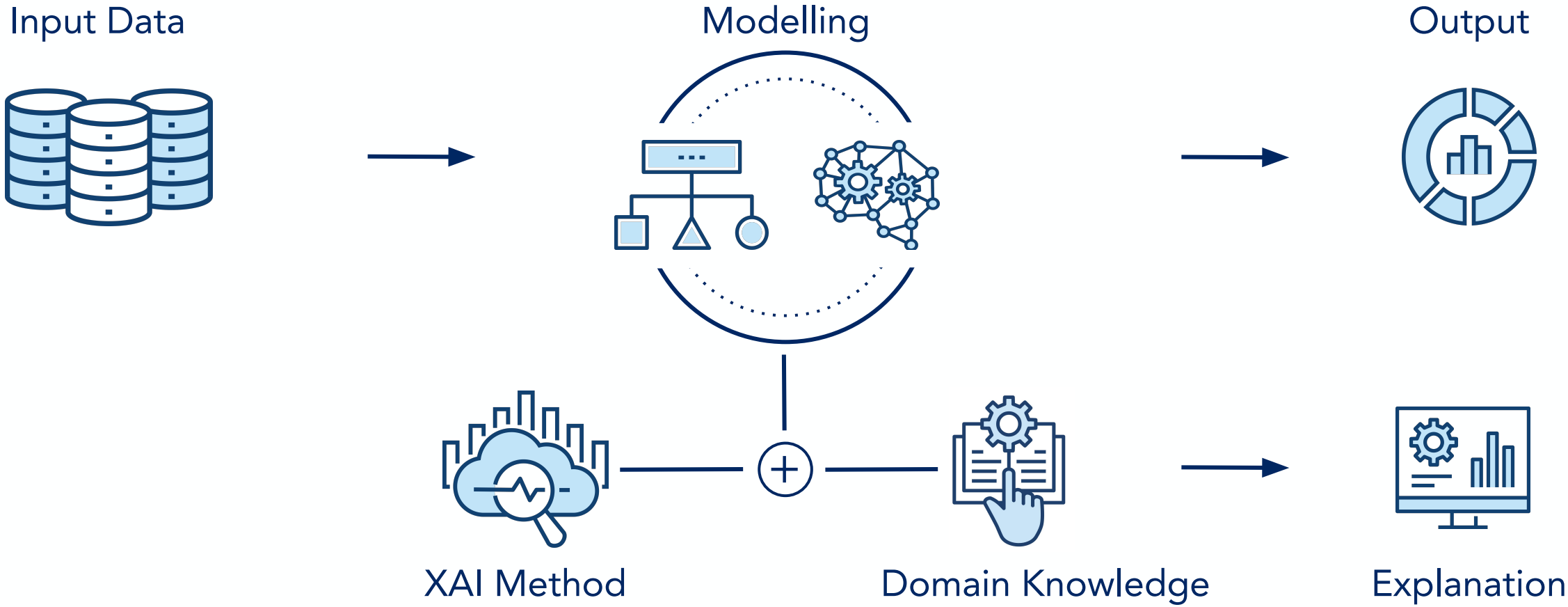
Introduction to eXplainable AI (XAI)

Integration of XAI into your Machine Learning workflow



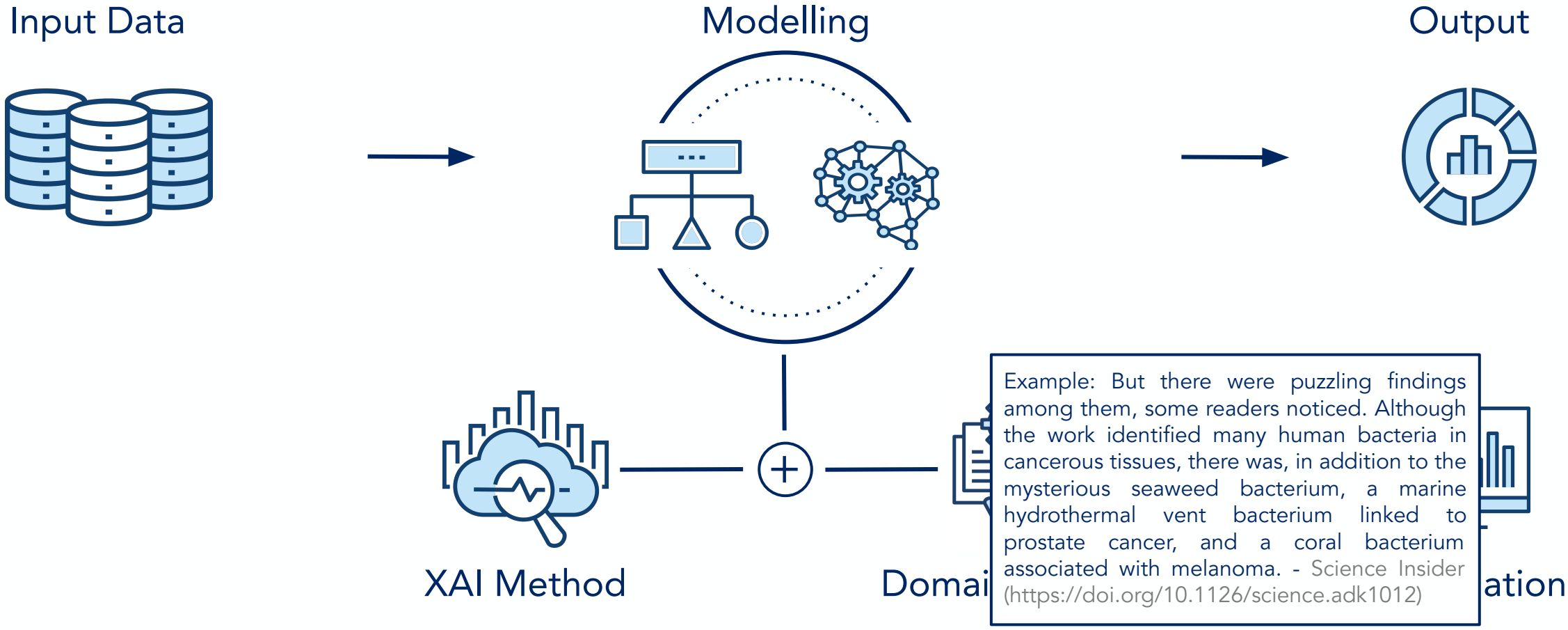
Introduction to eXplainable AI (XAI)

Integration of XAI into your Machine Learning workflow



Introduction to eXplainable AI (XAI)

Integration of XAI into your Machine Learning workflow



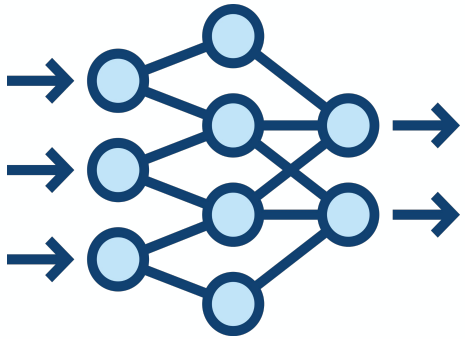
Introduction to eXplainable AI (XAI)

Integration of XAI into your Machine Learning workflow

Input Data



Modelling



Output

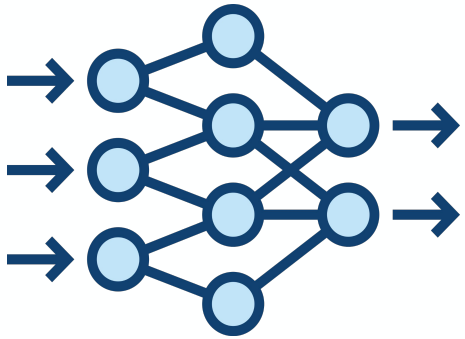
Introduction to eXplainable AI (XAI)

Integration of XAI into your Machine Learning workflow

Input Data



Modelling



Current explanation: This is a cat!

Output



Cat

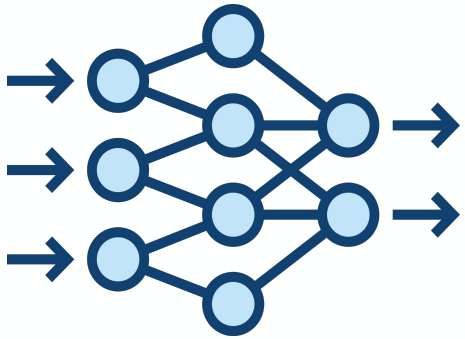
Introduction to eXplainable AI (XAI)

Integration of XAI into your Machine Learning workflo

Input Data



Modelling



Output

Current explanation: This is a cat!



Cat

XAI Method ⊕

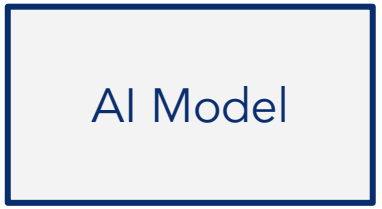
Domain Knowledge

XAI explanation:
- it has fur, whiskers, and claws
- it has this feature



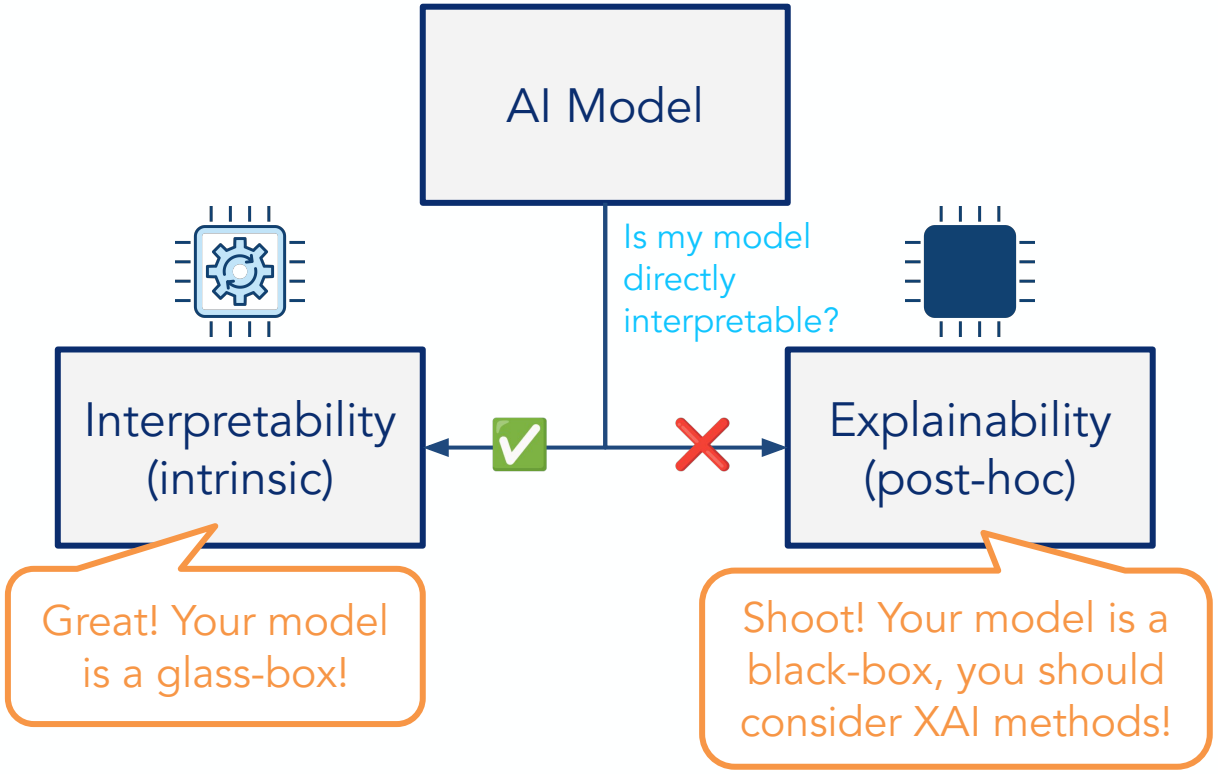
Introduction to eXplainable AI (XAI)

Integration of XAI into your Machine Learning workflow



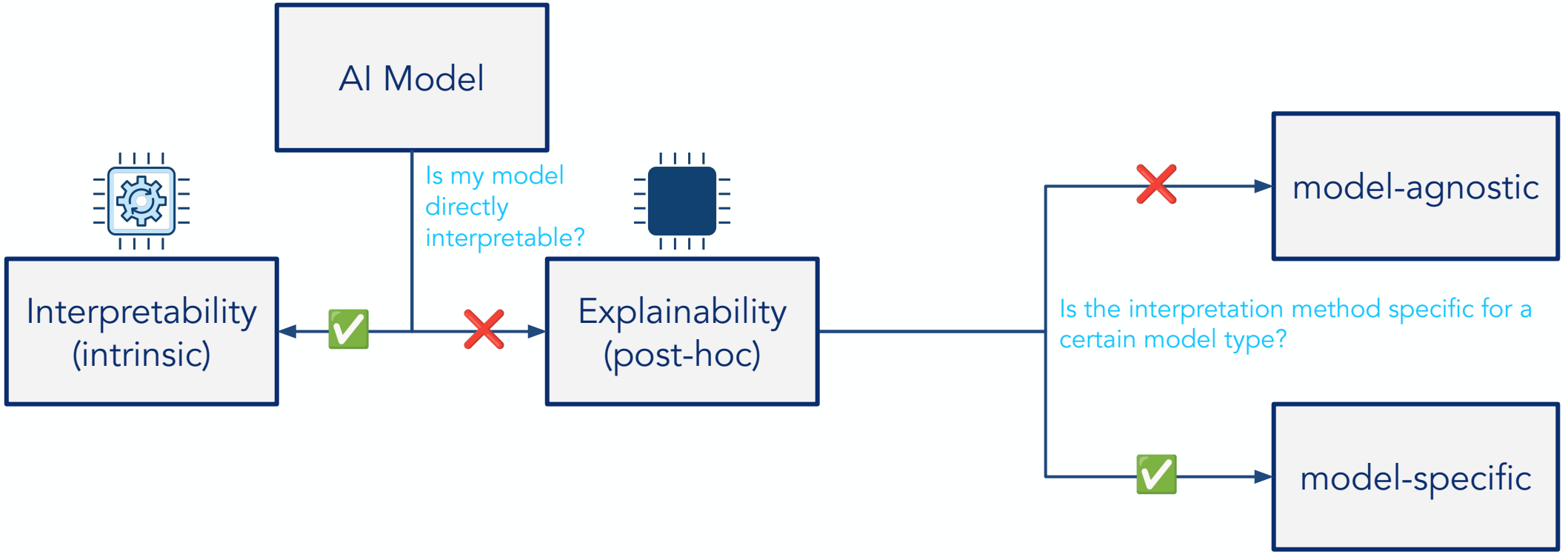
Introduction to eXplainable AI (XAI)

Integration of XAI into your Machine Learning workflow



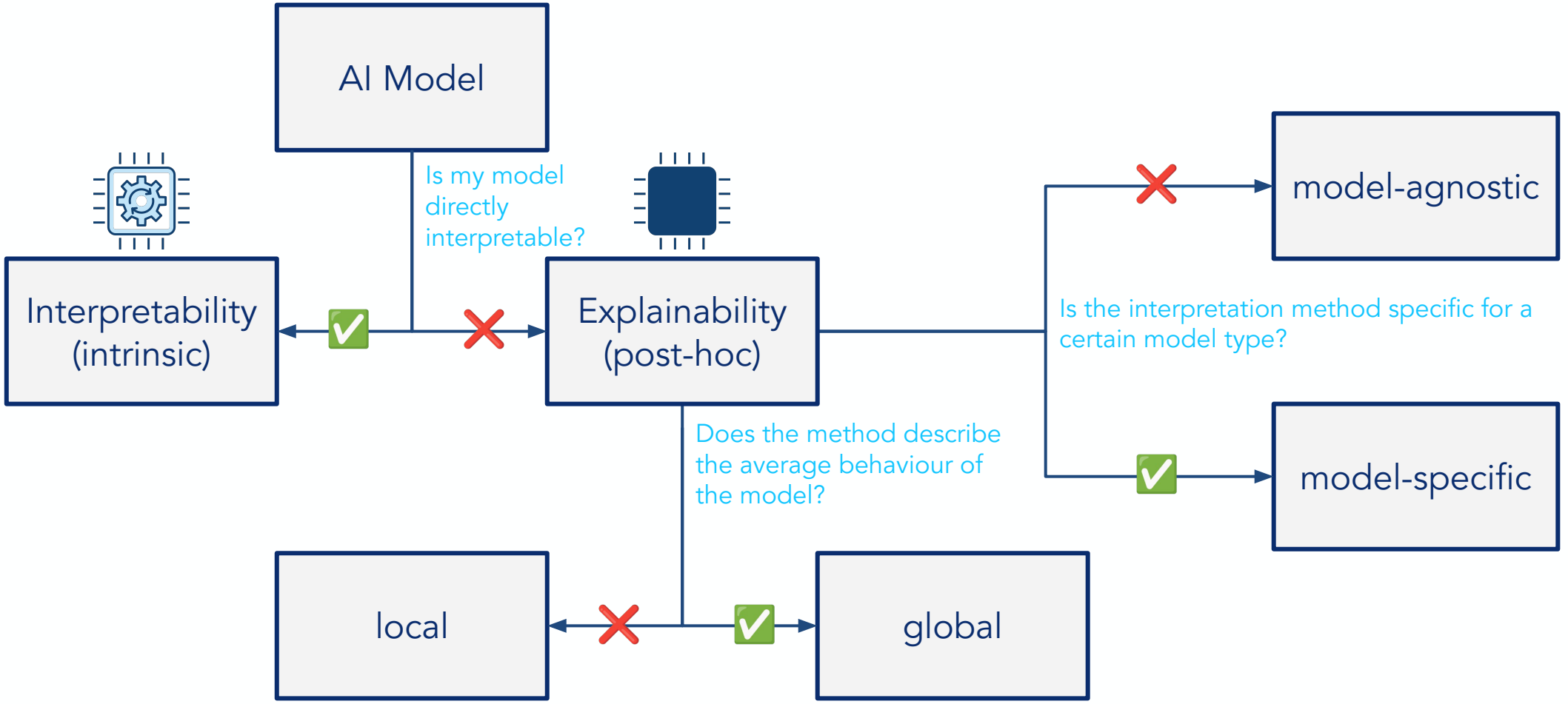
Introduction to eXplainable AI (XAI)

Integration of XAI into your Machine Learning workflow



Introduction to eXplainable AI (XAI)

Integration of XAI into your Machine Learning workflow



slido

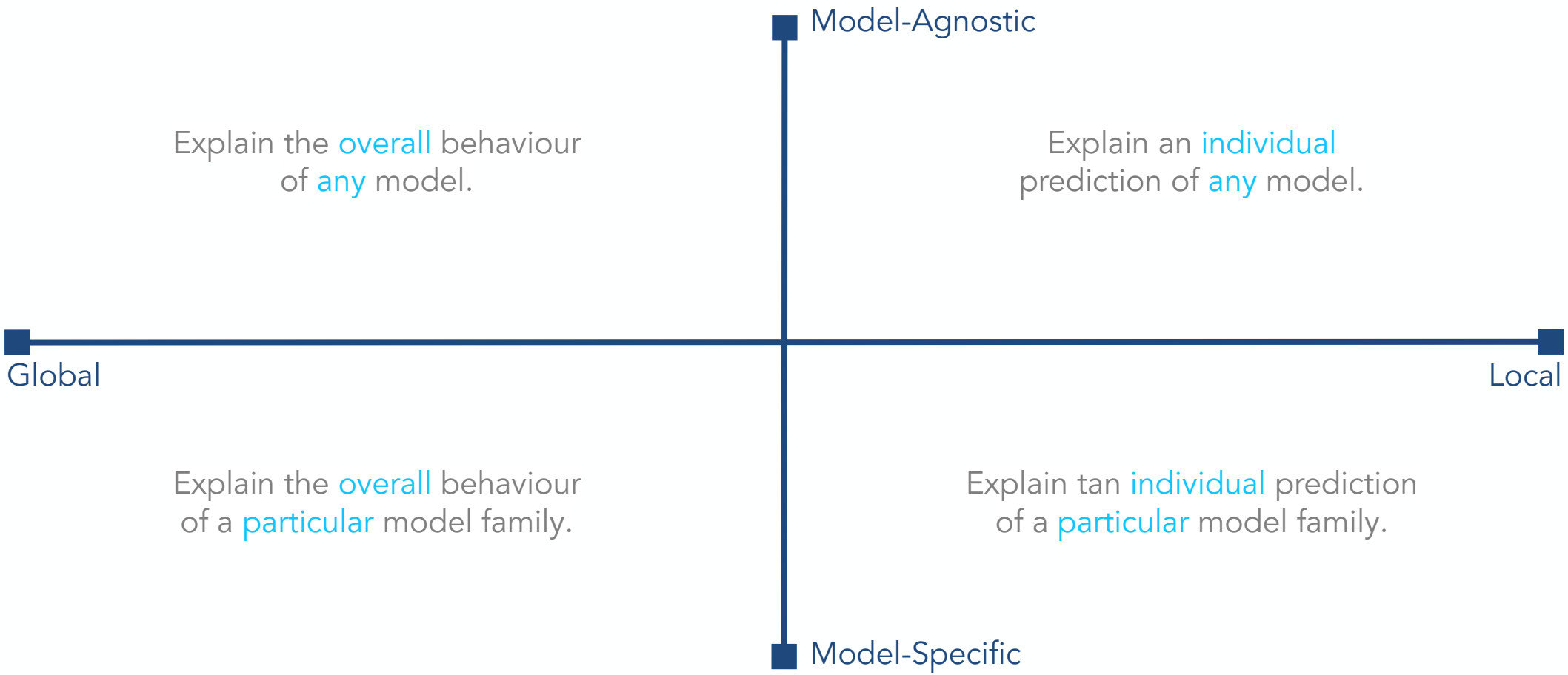


To understand what impact blood pressure has on the survival rate of patient John Doe in a Random Forest model, we need:

ⓘ Start presenting to display the poll results on this slide.

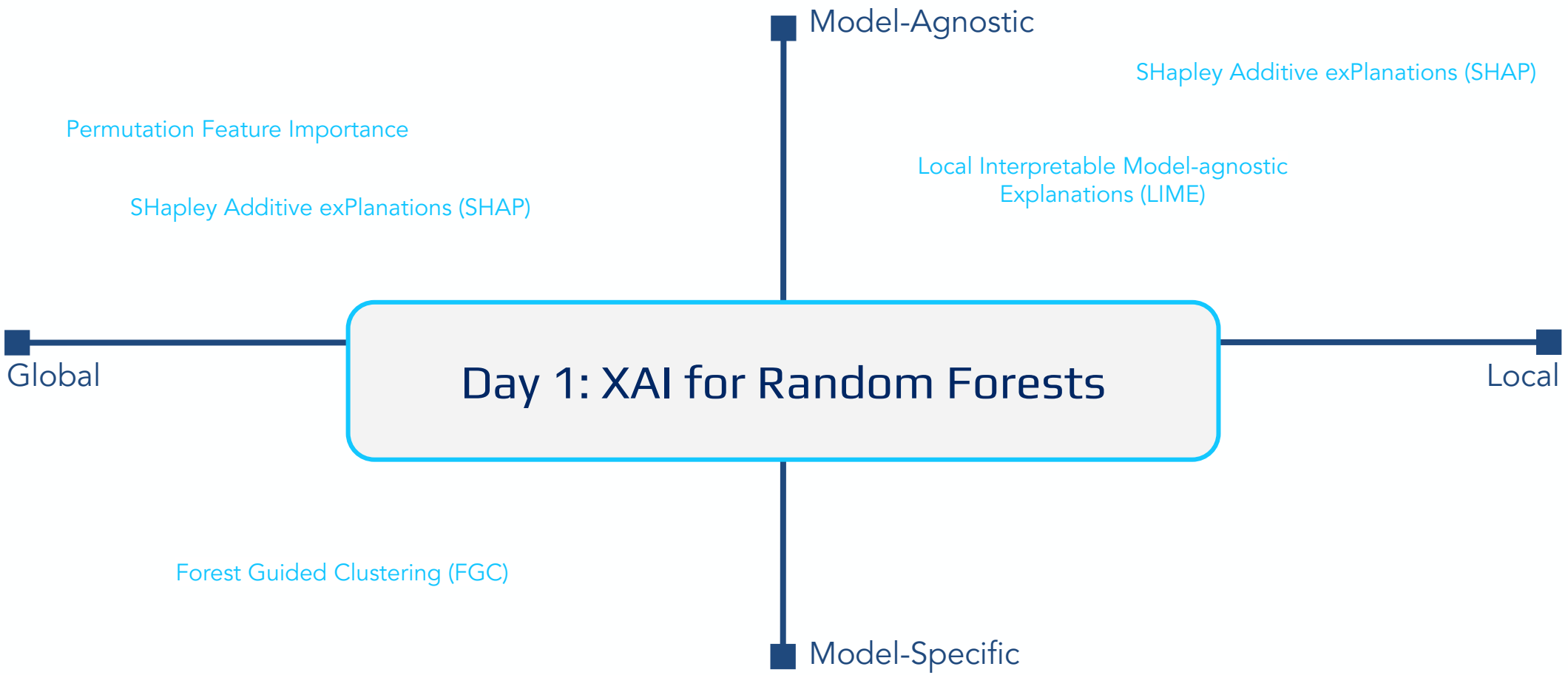
Introduction to eXplainable AI (XAI)

Integration of XAI into your Machine Learning workflow



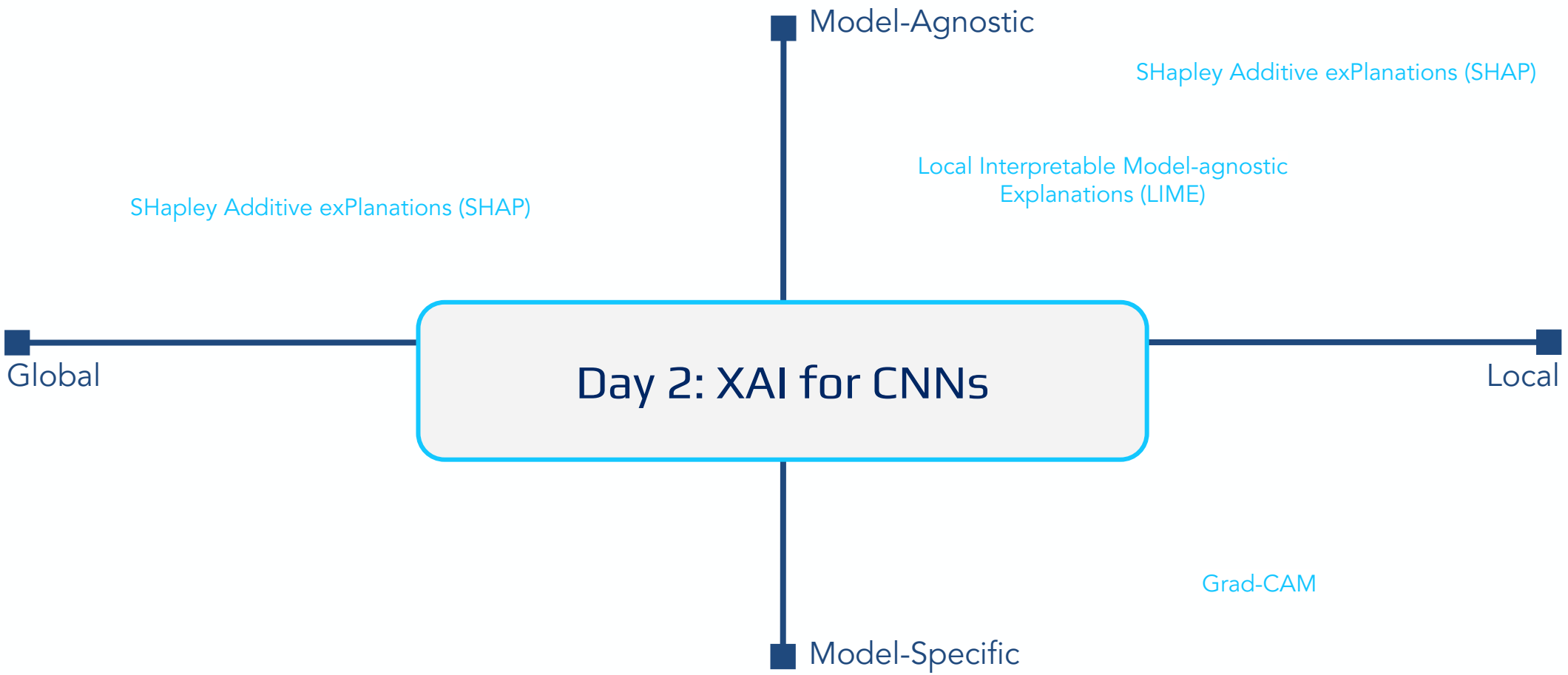
Introduction to eXplainable AI (XAI)

Integration of XAI into your Machine Learning workflow



Introduction to eXplainable AI (XAI)

Integration of XAI into your Machine Learning workflow



You will move now into separate breakout sessions with your tutors!

Agenda

Introduction to eXplainable AI (XAI)

Day 1: XAI for Random Forests

- Model-agnostic methods: Perm. Feature Importance, LIME, SHAP
- Model-specific methods: Forest-Guided Clustering
- Comparison of XAI methods for Random Forest models

Day 2: XAI for CNNs

- Model-agnostic methods: LIME, SHAP
- Model-specific methods: Grad-CAM
- Comparison of XAI methods for CNN models

Wrap-Up

GitHub Repository:
https://tinyurl.com/4y46r666

Agenda

Introduction to eXplainable AI (XAI)

Day 1: XAI for Random Forests

- Model-agnostic methods: Perm. Feature Importance, LIME, SHAP
- Model-specific methods: Forest-Guided Clustering
- Comparison of XAI methods for Random Forest models

Day 2: XAI for CNNs

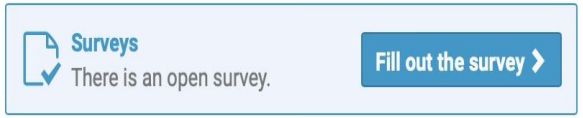
- Model-agnostic methods: LIME, SHAP
- Model-specific methods: Grad-CAM
- Comparison of XAI methods for CNN models

Wrap-Up

Wrap-Up

Survey

Survey:



Further Tutorials:

XAI for Transformers

Further Reading

